

MSAFE AI

Dashboard Working & Display Guide

A clear description of how the MSAFE AI dashboard works, what each section displays, and how mine-subsidence data moves from sensors to risk and early warning.

1. Purpose

MSAFE AI is an AI-enabled, low-cost, real-time mine subsidence monitoring, prediction and early-warning system. The dashboard converts mine data, physics-model output and surface sensor measurements into an operational view of the monitored area. It is intended to show the current condition, identify abnormal deformation, estimate progression, visualize affected/risk zones and generate early warnings.

2. Main Dashboard Hierarchy

Priority	Section	What it shows
1	Current Status	Overall risk state: NORMAL, WATCH, WARNING, CRITICAL or UNKNOWN, with confidence.
2	Key Metrics	Observed deformation, expected deformation, difference, confidence and sensor health.
3	Mine / GIS Map	Mine boundary, panel, sensor locations, affected/risk zone and predicted zone.
4	Model vs Observed	Expected deformation compared with observed deformation.
5	AI & Prediction	Anomaly level, progression state, prediction and important factors.
6	Alerts	Active and recent warnings/events.

3. End-to-End Pipeline

MINE DATA → PHYSICS MODEL → EXPECTED DEFORMATION → SENSOR NETWORK → OBSERVED DEFORMATION → MODEL vs OBSERVED → AI ANOMALY → PROGRESSION → RISK ASSESSMENT → GIS RISK ZONE → EARLY WARNING

Tilt, displacement, vibration, crack opening, temperature and node health belong inside the Sensor Network stage. Deformation rate, rate of change, affected area and persistence are analysis indicators. Confidence is displayed alongside risk, not as a separate pipeline stage.

4. Navigation

Group	Page	Purpose
MONITORING	Overview	Main operational dashboard.
	Live Monitoring	Live sensor readings and node/network condition.
	GIS Map	Spatial view of mine, sensors, affected areas and risk zones.
	Sensors	Individual sensor/node status.
ANALYSIS	Model vs Observed	Detailed expected-versus-observed analysis.
	AI & Prediction	Anomaly detection, progression and prediction.
EVENTS	Alerts	Current and previous alerts.
	History	Historical measurements and events.
SYSTEM	Settings	Application/system configuration.

5. Overview Dashboard

A. Current Status

This is the first visual priority. It communicates the current overall condition. Risk states are NORMAL, WATCH, WARNING, CRITICAL or UNKNOWN. Risk and confidence are shown separately.

B. Key Metrics

Metric	Example	Meaning
Current Risk	WATCH	Current system-level risk state.
Observed	42.6 mm	Observed deformation.
Expected	30.8 mm	Model-expected deformation.
Confidence	94%	Confidence in the current analysis/risk result.
Sensors	7/7	Reporting/healthy sensor nodes.

C. Mine / GIS Map

The GIS map is the main spatial visual. It shows the mine boundary and panel, sensor locations, current affected/risk areas and predicted zones when available. Optional layers should be controlled through a layer panel so the map remains readable.

6. Sensor Network

Sensor / indicator	Dashboard role
Tilt / inclination	Shows change in surface inclination.
Relative displacement	Shows local movement/deformation.
Vibration	Captures vibration/activity.
Crack opening	Tracks change in crack width/opening.
Temperature	Provides environmental/context information.
Battery / node health	Shows whether a node is functioning and reporting.
LoRa/network health	Shows connectivity and communication condition.

7. Model vs Observed

The dashboard compares deformation expected from the 2-D PIM physics model with deformation observed through the sensor network. The UI should use simple operational language rather than exposing mathematical formulas.

Displayed item	Example	Interpretation
Model Expected	30.8 mm	Expected deformation from the model.
Sensor Observed	42.6 mm	Observed deformation.
Difference	11.8 mm	Difference between expected and observed.
Deformation Rate	2.4 mm/h	How quickly deformation is changing.
Rate of Change	Increasing	Whether the rate is increasing/decreasing/stable.
Affected Area	420 m ²	Approximate affected/anomalous area.
Persistence	Sustained	Whether the abnormal condition continues.

8. AI & Prediction

Display	What it shows
AI Anomaly Level	Simple operational level such as Low, Medium or High.
Confidence	Confidence associated with the analysis.
Key factors	Model-vs-observed difference, deformation rate, crack change and sensor agreement.
Progression	Stable, Developing, Accelerating or Rapid.
Predicted zone	Area where continued deformation is predicted/considered relevant.

9. Risk Assessment & Early Warning

Risk is displayed as NORMAL, WATCH, WARNING, CRITICAL or UNKNOWN. Confidence is separate. A short reason explains the state, such as observed deformation being above model expectation, an increasing deformation rate, or agreement between multiple sensors.

10. Alerts & History

The Overview page should show only active/recent alerts. The dedicated Alerts page can contain severity, timestamp, location, reason and status. History is used to review earlier measurements, risk states and events.

11. Gateway & Communication

Surface nodes communicate through an adaptive LoRa mesh. The Raspberry Pi 5 edge gateway collects data, supports timestamp synchronization, monitors sensor/network health, stores data locally and can support local/offline processing before synchronization with the cloud dashboard.

12. Suggested Overview Layout

Area	Content
Top bar	MSAFE AI Mine status Connection/sync state User account
Row 1	Current Risk Observed Expected Confidence Sensors
Main visual	Large GIS/Mine Map with sensors, risk zone and predicted zone
Analysis row	Model vs Observed chart AI & Prediction summary
Bottom	Active Alerts Recent events / system health
Sidebar	Overview, Live Monitoring, GIS Map, Sensors, Model vs Observed, AI & Prediction, Alerts, History, Settings

13. Example Dashboard Reading

The map then identifies where the affected area is located. The Model vs Observed section shows whether measurements are diverging from the model. AI & Prediction summarizes anomaly level and progression. Alerts communicate whether an operator needs to act. In other words: what is happening → where it is happening → how it is progressing → what warning is currently required.

14. UI Design Rules

- Use a dark navy/charcoal interface with restrained cyan/blue accents.
- Use green for NORMAL, amber for WATCH, orange for WARNING, red for CRITICAL and gray for UNKNOWN.
- Use Inter or a standard system font. Avoid futuristic/pixel/decorative fonts.
- Keep the interface responsive for laptop, tablet and mobile.
- Keep the desktop sidebar fully visible; use compact/drawer navigation on smaller screens.
- Avoid excessive gradients, glow effects and oversized rounded cards.
- Keep formulas and implementation details internal. Display operational meaning in plain language.
- Keep Overview focused. Put detailed information on dedicated pages.

15. System Architecture Summary

Mine/site data → 2-D physics model → expected deformation → surface sensor nodes → adaptive LoRa mesh → Raspberry Pi 5 edge gateway → observed deformation → model-vs-observed analysis → multi-sensor fusion → AI anomaly detection → progression prediction → risk + confidence → GIS risk zone → early warning/dashboard.

16. Prototype Context

The proposed prototype uses distributed low-cost surface sensor nodes, LoRa communication, a Raspberry Pi 5 edge gateway and a web/cloud dashboard. A controlled physical demonstration can show how deformation changes are sensed, transmitted, analyzed and converted into a changing dashboard risk state.